

Sample Paper 2 — Class 5 Mathematics Olympiad

Pattern: SOF IMO Level 1 — 50 questions, 60 marks, 60 minutes, no negative marking. Set a timer before you start.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Apple B) Banana C) Potato D) Mango
- Q2.** Fish is to Water as Bird is to _____. A) Nest B) Sky C) Tree D) Cage
- Q3.** If each letter of a word is moved 1 step forward in the alphabet, BAT becomes CBU. Using the same rule, how is DOG written? A) EPH B) EPG C) DPH D) EOH
- Q4.** Complete the series: 5, 10, 20, 40, 80, ____ A) 120 B) 140 C) 160 D) 180
- Q5.** In a row of 25 students, Vikram is 9th from the left. What is his position from the right? A) 16th B) 17th C) 18th D) 15th
- Q6.** Facing North, Rohit turns 90° clockwise, then another 90° clockwise. Which direction does he face now? A) East B) South C) West D) North
- Q7.** Arrange in a logical growth order: 1. Child 2. Adult 3. Infant 4. Teenager A) 3, 1, 4, 2 B) 1, 2, 3, 4 C) 3, 4, 1, 2 D) 1, 3, 4, 2
- Q8.** Find the odd one out: A) 31 B) 37 C) 39 D) 41
- Q9.** If today is Monday, what day will it be after 26 days? A) Friday B) Saturday C) Sunday D) Thursday
- Q10.** Pointing to a boy, Meena (who has no sisters) said, "He is the only son of my father's only daughter." How is the boy related to Meena? A) Brother B) Son C) Nephew D) Cousin
- Q11.** Find the missing number: 3, 7, 15, 31, ____ A) 59 B) 61 C) 63 D) 65
- Q12.** In a certain code, HOUSE is written as JQWUG (each letter moved 2 steps forward). How is MOUSE written in the same code? A) OQWUG B) OPWUG C) OQWUH D) PQWUG
- Q13.** Find the odd one out: A) Cone B) Cylinder C) Triangle D) Sphere
- Q14.** P is older than Q. R is younger than Q. S is older than P. Who is the oldest? A) P B) Q C) R D) S
- Q15.** Complete the series: 2, 5, 10, 17, 26, ____ A) 35 B) 36 C) 37 D) 38

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

- Q16.** $56,789 + 34,567 = ?$ A) 91,356 B) 91,256 C) 90,356 D) 91,456
- Q17.** $90,003 - 45,678 = ?$ A) 44,325 B) 44,225 C) 44,425 D) 45,325
- Q18.** $4,567 \times 9 = ?$ A) 41,103 B) 41,003 C) 41,203 D) 40,103
- Q19.** $86,912 \div 8 = ?$ A) 10,864 B) 10,764 C) 10,964 D) 10,854
- Q20.** Find the HCF of 32 and 48. A) 8 B) 12 C) 16 D) 24
- Q21.** Find the LCM of 9 and 12. A) 36 B) 72 C) 18 D) 108
- Q22.** Which fraction is equivalent to $\frac{4}{5}$? A) $\frac{8}{10}$ B) $\frac{5}{4}$ C) $\frac{4}{10}$ D) $\frac{8}{5}$
- Q23.** $\frac{3}{8} + \frac{2}{8} + \frac{1}{8} = ?$ A) $\frac{3}{4}$ B) $\frac{6}{24}$ C) $\frac{1}{2}$ D) 1
- Q24.** Convert 0.45 to a fraction in simplest form. A) $\frac{9}{20}$ B) $\frac{4}{5}$ C) $\frac{9}{10}$ D) $\frac{4}{9}$
- Q25.** What is 35% of 240? A) 74 B) 80 C) 84 D) 90
- Q26.** Express $\frac{7}{10}$ as a percentage. A) 17% B) 70% C) 7% D) 71%
- Q27.** Two numbers are in the ratio 3:5. If the smaller is 18, find the larger. A) 25 B) 28 C) 30 D) 35
- Q28.** The cost price of an item is ₹320 and the selling price is ₹400. Find the profit percentage. A) 20% B) 25% C) 30% D) 35%
- Q29.** Find the simple interest on ₹4,000 at 6% per annum for 5 years. A) ₹1,000 B) ₹1,100 C) ₹1,200 D) ₹1,300
- Q30.** Convert 5 kg 80 g into grams. A) 5,080 g B) 5,008 g C) 5,800 g D) 580 g
- Q31.** Convert 2 hours 15 minutes into minutes. A) 125 min B) 130 min C) 135 min D) 140 min
- Q32.** Find the perimeter of a rectangle with length 19 cm and breadth 11 cm. A) 30 cm B) 60 cm C) 120 cm D) 209 cm
- Q33.** Find the area of a square with side 14 cm. A) 186 sq cm B) 196 sq cm C) 56 sq cm D) 206 sq cm
- Q34.** How many lines of symmetry does a (non-equilateral) isosceles triangle have? A) 0 B) 1 C) 2 D) 3
- Q35.** How many right angles make three-quarters of a full turn? A) 2 B) 3 C) 4 D) 6

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

- Q36.** A factory produces 3,200 toys each day. How many toys does it produce in 25 days? A) 78,000 B) 80,000 C) 82,000 D) 84,000
- Q37.** Arjun had ₹8,000. He spent ₹2,350 and then ₹1,875. How much money does he have left? A) ₹3,775 B) ₹3,875 C) ₹3,675 D) ₹3,975

- Q38.** A tank holds 540 L. If $\frac{2}{3}$ of it is used, how much water is left? A) 160 L B) 180 L C) 200 L D) 220 L
- Q39.** 18 kg of sugar costs ₹900. Find the cost of 1 kg. A) ₹45 B) ₹48 C) ₹50 D) ₹52
- Q40.** A school has 9 classrooms with 36 students in each. How many students are there in total? A) 314 B) 324 C) 334 D) 344
- Q41.** Reema deposited ₹3,000 at 7% simple interest per year. How much interest will she earn in 4 years? A) ₹800 B) ₹820 C) ₹840 D) ₹860
- Q42.** A trader bought goods for ₹1,800 and sold them for ₹2,070. Find the profit percentage. A) 10% B) 15% C) 20% D) 25%
- Q43.** A train travels at 75 km/h. How far will it travel in 1 hour 40 minutes? A) 115 km B) 120 km C) 125 km D) 130 km
- Q44.** A recipe needs $\frac{2}{3}$ cup of flour. How much flour is needed for 6 such recipes? A) 3 cups B) 4 cups C) 5 cups D) 6 cups
- Q45.** In a survey of 240 students, 45% like football. How many students like football? A) 98 B) 104 C) 108 D) 112

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

- Q46.** A number is decreased by 25% to get 225. Find the original number. A) 280 B) 290 C) 300 D) 310
- Q47.** The sum of three consecutive multiples of 5 is 90. Find the largest. A) 30 B) 35 C) 40 D) 45
- Q48.** The simple interest on a sum for 4 years at 8% per annum is ₹3,200. Find the sum. A) ₹8,000 B) ₹9,000 C) ₹10,000 D) ₹11,000
- Q49.** Priya had some money. She spent $\frac{1}{3}$ of it on a dress and $\frac{1}{4}$ of the remainder on shoes. She is left with ₹900. How much money did she have at first? A) ₹1,500 B) ₹1,600 C) ₹1,800 D) ₹2,000
- Q50.** The ratio of the ages of two friends is 2:3. After 5 years, the ratio becomes 3:4. Find the elder friend's present age. A) 12 B) 15 C) 18 D) 20

Answer key: [sample-paper-02-answer-key.md](#)
